The eighteen 2-connected cores

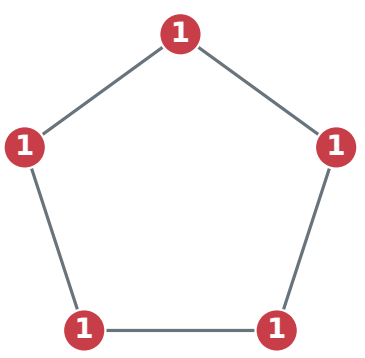
Every 2-connected graph with exactly five harmonic boundary vertices appears here.

Boundary: positive score Interior: nonpositive score

Each node is labelled by its exact stability $\beta_G(u) = \max_v \sum_{w \sim u} [d(u,v) - d(w,v)]$.

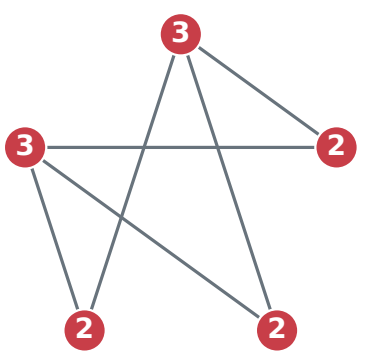
EIGHT CORES ON FIVE VERTICES

5N1
5 vertices · 5 edges



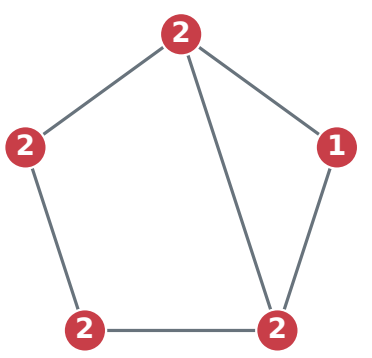
graph6: Dhc

5N2
5 vertices · 6 edges



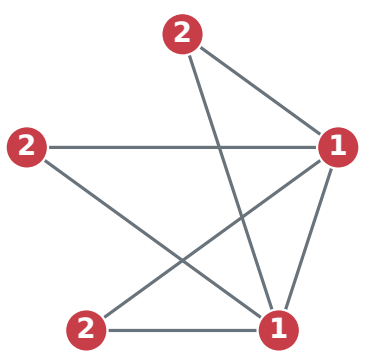
graph6: D]o

5N3
5 vertices · 6 edges



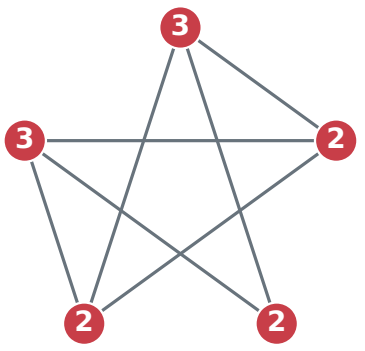
graph6: Dlc

5N4
5 vertices · 7 edges



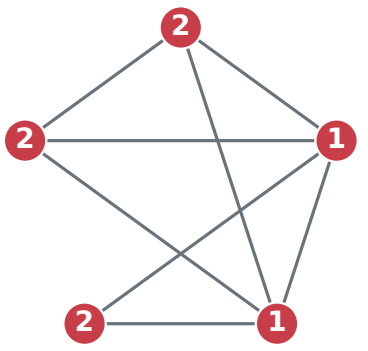
graph6: DF{

5N5
5 vertices · 7 edges



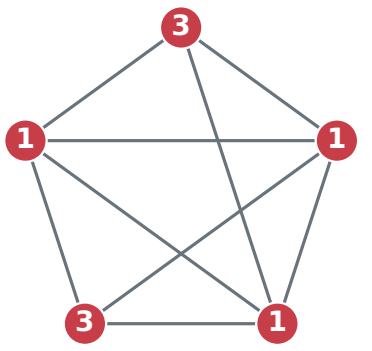
graph6: D]w

5N6
5 vertices · 8 edges



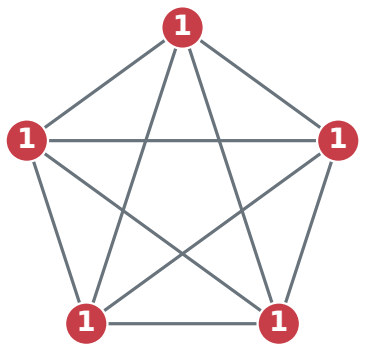
graph6: Df{

5N7
5 vertices · 9 edges



graph6: Dn{

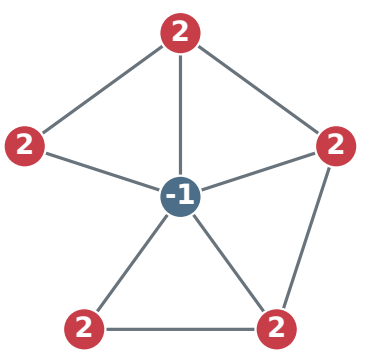
5N8
5 vertices · 10 edges



graph6: D~{

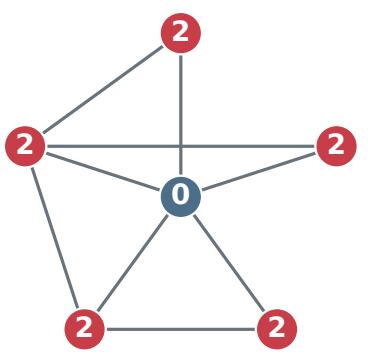
TEN CORES ON SIX VERTICES

6N1
6 vertices · 9 edges



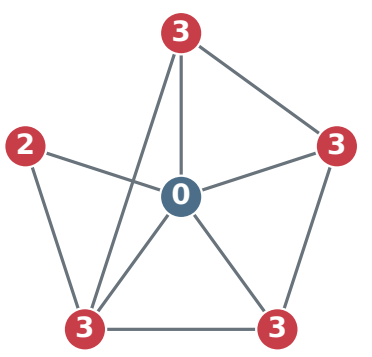
graph6: EyVG

6N2
6 vertices · 9 edges



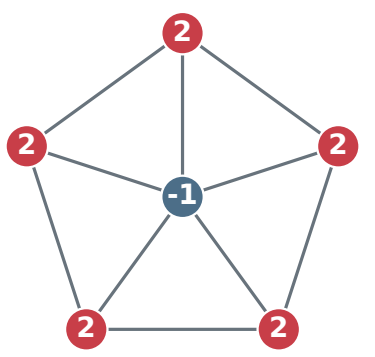
graph6: E|e_

6N3
6 vertices · 10 edges



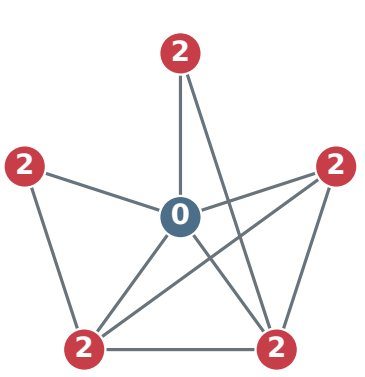
graph6: E^Mg

6N4
6 vertices · 10 edges



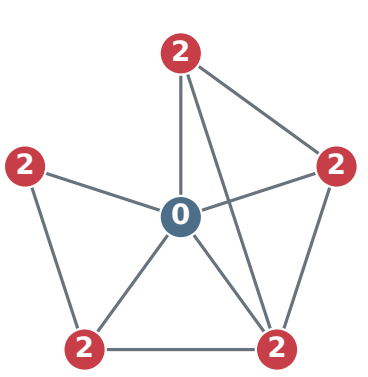
graph6: Ehfw

6N5
6 vertices · 10 edges



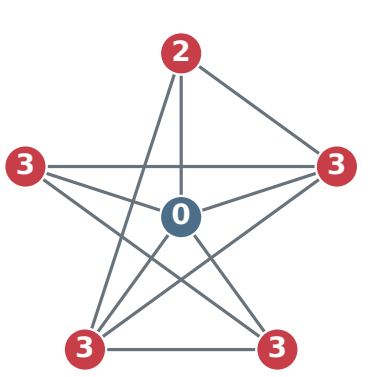
graph6: EjtW

6N6
6 vertices · 10 edges



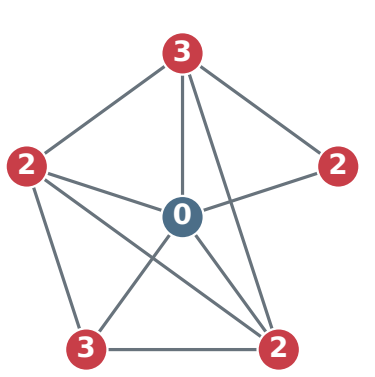
graph6: EjvG

6N7
6 vertices · 11 edges



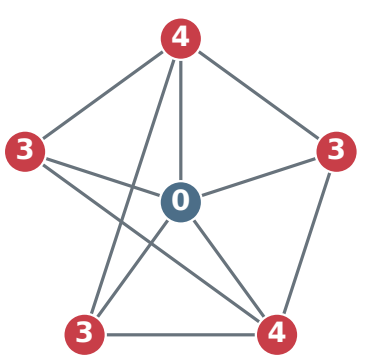
graph6: ER~g

6N8
6 vertices · 11 edges



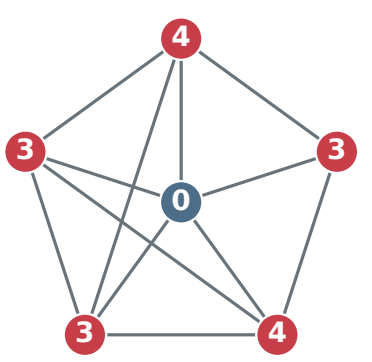
graph6: En}G

6N9
6 vertices · 11 edges



graph6: E}^G

6N10
6 vertices · 12 edges



graph6: E~^G

Exact integer BFS verification · all eighteen graphs are pairwise catalogued cores · graph6 codes preserve the source vertex labels